SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 0607 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

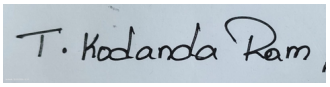
QUESTIONS: 15 **Total Marks: 25**

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I T . Kodanda Ram (192524162) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C? (5 Marks)

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis. (5Marks)

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

a. Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
b. Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.
ii) Suggest two effective techniques to reduce overfitting in this model.
iii) Discuss how overfitting affects the model's generalization ability on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)
(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
(c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
(h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%
i) Identify the epoch at which the model starts overfitting.
ii) Explain why validation accuracy decreases despite training accuracy increasing.
iii) Suggest two methods to improve validation performance.

6. A machine learning model shows the following loss values during training: (5 Marks)

(a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
(b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3
(c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
(d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
(e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
(f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5
(g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8
(h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1
i) At which epoch does the model achieve optimal generalization?
ii) Identify signs of overfitting from the data.
iii) Recommend two regularization techniques and explain their impact.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires). (5 Marks)

a. Front identified as Front: 119
b. Front identifies as Back: 06
c. Back identified as Back: 120
d. Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

a. Diseased identified as Diseased: 138
b. Diseased identified as Healthy: 12
c. Healthy identified as Healthy: 135
d. Healthy identified as Diseased: 15

Compute: **Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.**

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective):
(5 Marks)

- Defective identified as Defective: 2 30
- Defective identified as Non-Defective: 20
- Non-Defective identified as Non-Defective: 225
- Non-Defective identified as Defective: 25

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated.
(5 Marks)

12. A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective. (5 Marks)

13. In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150.The dataset contains 1000 samples per class, and 20% of the data is used for testing.
(5 Marks)
Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data.
Calculate: Accuracy Precision Sensitivity Specificity. (5 Marks)

15. A deep learning model is trained for a classification task, and the following loss values are observed: (5 Marks)

i. **Epoch 1: Training Loss = 2.8, Validation Loss = 2.6**
ii. **Epoch 2: Training Loss = 2.4, Validation Loss = 2.2**
iii. **Epoch 3: Training Loss = 2.1, Validation Loss = 1.9**
iv. **Epoch 4: Training Loss = 1.9, Validation Loss = 1.8**
v. **Epoch 5: Training Loss = 1.7, Validation Loss = 1.9**
vi. **Epoch 6: Training Loss = 1.5, Validation Loss = 2.1**
vii. **Epoch 7: Training Loss = 1.3, Validation Loss = 2.4**
viii. **Epoch 8: Training Loss = 1.2, Validation Loss = 2.7**

Answer the following:
a. **At which epoch does the model achieve the best generalization performance?**
b. **At which point does overfitting begin? Justify your answer.**
c. **Suggest two techniques to reduce overfitting and briefly explain how they help.**

Question 1

Full Question: A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C? (5 Marks)

Step 1 — Prior probabilities

Let P(A)=x, P(C)=x and P(B)=2x.
x + 2x + x = 1
4x = 1
P(A) = 1/4
P(B) = 1/2
P(C) = 1/4

Step 2 — Blue probabilities

P(Blue|A) = 4/12
= **1/3**

P(Blue|B) = 7/15

P(Blue|C) = 2/15

Step 3 — Total probability of Blue

P(Blue) = P(A)P(Blue|A) + P(B)P(Blue|B) + P(C)P(Blue|C)
= (1/4)(1/3) + (1/2)(7/15) + (1/4)(2/15)
= 1/12 + 7/30 + 1/30
= 5/60 + 14/60 + 2/60
= **21/60**
= **7/20**

Step 4 — Bayes' theorem

P(C|Blue) = P(C)P(Blue|C) / P(Blue)
= [(1/4)(2/15)] / (7/20)
= (1/30) × (20/7)
= 20/210
= **2/21**

Final Answer

P(C|Blue) = 2/21 = 0.0952 = 9.52%.

Question 2

Full Question: Aparking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black

cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)

Step 1 — Prior probabilities

P(A) = 1/4
P(B) = 1/2
P(C) = 1/4

Step 2 — Blue probabilities

P(Blue|A) = 3/10
P(Blue|B) = 6/15
= 2/5

P(Blue|C) = 2/15

Step 3 — Total probability

P(Blue) = (1/4)(3/10) + (1/2)(2/5) + (1/4)(2/15)
= 3/40 + 1/5 + 1/30
= 9/120 + 24/120 + 4/120
= **37/120**

Step 4 — Bayes' theorem

P(C|Blue) = [(1/4)(2/15)] / (37/120)
= (1/30) × (120/37)
= 120/1110
= **4/37**

Final Answer

P(C|Blue) = 4/37 = 10.81%.

Question 3

Full Question: In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis. (5 Marks)

Step 1 — Total and test data

Total samples = 1800 + 1800
= **3600**

Test samples = 20/100 × 3600
= **720**

Positive test samples = 20/100 × 1800
= **360**

Step 2 — Find FN

TP + FN = 360
210 + FN = 360
FN = 360 – 210
FN = 150

Step 3 — Accuracy

Accuracy = (TP + TN) / Total
= (210 + 190) / 720
= 400 / 720
= **55.56%**

Step 4 — Precision

Precision = TP / (TP + FP)
= 210 / (210 + 30)
= 210 / 240
= **87.50%**

Step 5 — Sensitivity / Recall

Sensitivity = TP / (TP + FN)
= 210 / (210 + 150)
= 210 / 360
= **58.33%**

Step 6 — Specificity

Specificity = TN / (TN + FP)
= 190 / (190 + 30)

= 190 / 220
= **86.36%**

Final Answer / Interpretation

FN = **150**
Accuracy = **55.56%**
Precision = **87.50%**
Sensitivity = **58.33%**
Specificity = **86.36%**

The precision and specificity are relatively high, but the sensitivity is lower, meaning some positive cases are missed.

Question 4

Full Question: Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4; (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2; (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0; (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1; (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3; (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6; (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9; (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2. i) Identify the epoch at which the model begins to overfit. ii) Suggest two effective techniques to reduce overfitting in this model. iii) Discuss how overfitting affects the model's generalization ability on unseen data. (5 Marks)

Step 1 — Observe validation loss

Epoch 1→ 2.4
Epoch 2→ 2.2
Epoch 3→ 2.0
Epoch 4→ 2.1
Epoch 5→ 2.3
Epoch 6→ 2.6
Epoch 7→ 2.9
Epoch 8 → 3.2

Step 2 — Identify overfitting

Minimum validation loss = **2.0 at Epoch 3**.
At Epoch 4:
Training loss = 2.0 → decreases
Validation loss = 2.1 → increases
Therefore, **overfitting begins at Epoch 4**.

Step 3 — Two techniques

1. **Early stopping:** stop training when validation loss starts increasing.
2. **Dropout / L2 regularization:** controls model complexity and reduces dependence on training-specific patterns.

Step 4 — Effect on generalization

Training loss continues decreasing, but validation loss increases.
This means the model learns training-specific patterns/noise.
Therefore, its performance on unseen data becomes poorer.
Conclusion: Overfitting reduces generalization ability.

Question 5

Full Question: Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%; (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%; (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%; (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%; (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%; (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%; (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%; (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%. i) Identify the epoch at which the model starts overfitting. ii) Explain why validation accuracy decreases despite training accuracy increasing. iii) Suggest two methods to improve validation performance. (5 Marks)

Step 1 — Observe validation accuracy

Epoch 1→ 58%
Epoch 2 → 63%

Epoch 3→ 68%
Epoch 4→ 72%
Epoch 5→ **73%**
Epoch 6→ 72%
Epoch 7→ 70%
Epoch 8 → 68%

Step 2—Identify overfitting

Highest validation accuracy = **73% at Epoch 5**.
At Epoch 6:
Training accuracy increases: 80% → 85%
Validation accuracy decreases: 73% → 72%
Therefore, **overfitting starts at Epoch 6**.

Step 3 — Reason

The model becomes too specialized to the training data.
Training accuracy ↑
Validation accuracy ↓
This indicates poor generalization to unseen data.

Step 4 — Two methods

1. **Early stopping**.
2. **Dropout / L2 regularization**.

Question 6

Full Question: A machine learning model shows the following loss values during training: (a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7; (b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3; (c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1; (d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0; (e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2; (f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5; (g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8; (h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1. i) At which epoch does the model achieve optimal generalization? ii) Identify signs of overfitting from the data. iii) Recommend two regularization techniques and explain their impact. (5 Marks)

Step 1 — Find minimum validation loss

Validation loss:
2.7 → 2.3 → 2.1 → **2.0** → 2.2 → 2.5 → 2.8 → 3.1
Minimum = **2.0 at Epoch 4**.

Step 2 — Optimal generalization

Optimal generalization occurs at Epoch 4.

Step 3 — Signs of overfitting

After Epoch 4:
Training loss: 1.9
Validation loss: 2.2 → 1.7 → 1.6 → 1.4 ↓
→ 2.5 → 2.8 → 3.1 ↑
Therefore, the model is overfitting.

Step 4 — Two techniques

L2 regularization: penalizes large weights and encourages a simpler model.
Dropout: randomly deactivates units during training and reduces over-dependence on training examples.

Question 7

Full Question: In an industrial process, an AI robot is assigned to segregate the front and back tires. Let the number of tires are 250 (125 front and 125 back tires). (5 Marks) a. Front identified as Front: 119 b. Front identifies as Back: 06 c. Back identified as Back: 120 d. Back identified as Front: 05. Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Step 1 — Confusion matrix

Take Front as positive.
TP = 119
FN = 6
TN = 120
FP = 5

Total = 119 + 6 + 120 + 5
= **250**

Step 2 — Accuracy

Accuracy = (TP + TN) / Total
= (119 + 120) / 250
= 239 / 250
= **95.60%**

Step 3 — Precision

Precision = TP / (TP + FP)
= 119 / (119 + 5)
= 119 / 124
= **95.97%**

Step 4 — Sensitivity

Sensitivity = TP / (TP + FN)
= 119 / (119 + 6)
= 119 / 125
= **95.20%**

Step 5 — Specificity

Specificity = TN / (TN + FP)
= 120 / (120 + 5)
= 120 / 125
= **96.00%**

Step 6 — F1-Score

F1 = 2TP / (2TP + FP + FN)
= 2(119) / [2(119) + 5 + 6]
= 238 / 249
= **95.58%**

Final Answer

Accuracy = **95.60%**
Precision = **95.97%**
Sensitivity = **95.20%**
Specificity = **96.00%**
F1-Score = **95.58%**

Question 8

Full Question: In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks) a. Diseased identified as Diseased: 138 b. Diseased identified as Healthy: 12 c. Healthy identified as Healthy: 135 d. Healthy identified as Diseased: 15. Compute: Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

Step 1 — Confusion matrix

Diseased = positive.
TP = 138
FN = 12
TN = 135
FP = 15
Total = 300

Step 2 — Accuracy

Accuracy = (TP + TN) / Total
= (138 + 135) / 300
= 273 / 300
= **91.00%**

Step 3 — Precision

Precision = TP / (TP + FP)
= 138 / (138 + 15)
= 138 / 153
= **90.20%**

Step 4 — Sensitivity

Sensitivity = TP / (TP + FN)
= 138 / (138 + 12)
= 138 / 150
= **92.00%**

Step 5 — Specificity

Specificity = TN / (TN + FP)
= 135 / (135 + 15)
= 135 / 150
= **90.00%**

Step 6 — F1-Score

F1 = 2TP / (2TP + FP + FN)
= 276 / (276 + 15 + 12)
= 276 / 303
= **91.09%**

Question 9

Full Question: In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks) Spam identified as Spam: 180; Spam identified as Not Spam: 20; Not Spam identified as Not Spam: 170; Not Spam identified as Spam: 30. Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Step 1 — Confusion matrix

Spam = positive.
TP = 180
FN = 20
TN = 170
FP = 30
Total = 400

Step 2 — Accuracy

Accuracy = (180 + 170) / 400
= 350 / 400
= **87.50%**

Step 3 — Precision

Precision = 180 / (180 + 30)
= 180 / 210
= **85.71%**

Step 4 — Sensitivity

Sensitivity = 180 / (180 + 20)
= 180 / 200
= **90.00%**

Step 5 — Specificity

Specificity = 170 / (170 + 30)
= 170 / 200
= **85.00%**

Step 6 — F1-Score

F1 = 2(180) / [2(180) + 30 + 20]
= 360 / 410
= **87.80%**

Question 10

Full Question: In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective): (5 Marks) Defective identified as Defective: 2 30; Defective identified as Non-Defective: 20; Non-Defective identified as Non-Defective: 225; Non-Defective identified as Defective: 25. Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Step 1 — Source question note

The uploaded question displays “2 30” for “Defective identified as Defective.” Since the question states 250 defective + 250 non-defective = 500, and the other entries are 20, 225 and 25, the internally consistent interpretation is **TP = 230**. This assumption is shown explicitly.

Step 2 — Confusion matrix

TP = 230
FN = 20
TN = 225
FP = 25
Total = 230 + 20 + 225 + 25
= **500**

Step 3 — Accuracy

Accuracy = (230 + 225) / 500
= 455 / 500
= **91.00%**

Step 4 — Precision

Precision = 230 / (230 + 25)
= 230 / 255
= **90.20%**

Step 5 — Sensitivity

Sensitivity = 230 / (230 + 20)
= 230 / 250
= **92.00%**

Step 6 — Specificity

Specificity = 225 / (225 + 25)
= 225 / 250
= **90.00%**

Step 7 — F1-Score

F1 = 2(230) / [2(230) + 25 + 20]
= 460 / 505
= **91.09%**

Question 11

Full Question: In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated. (5 Marks)

Step 1 — Prior probabilities

P(X) = 1/3
P(Y) = 2/3

Step 2 — Conditional probabilities

P(A|X) = 30 / 50
= **3/5**

P(A|Y) = 20 / 60
= **1/3**

Step 3 — Total probability

P(A) = (1/3)(3/5) + (2/3)(1/3)
= 1/5 + 2/9
= 9/45 + 10/45

= **19/45**

Step 4 — Bayes' theorem

P(Y|A) = [(2/3)(1/3)] / (19/45)
= (2/9) × (45/19)
= 90/171
= **10/19**

Final Answer

P(Y|A) = 10/19 = 52.63%.

Question 12

Full Question: A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective. (5 Marks)

Step 1 — Prior probabilities

P(A) = 3/4
P(B) = 1/4

Step 2 — Conditional probabilities

P(D|A) = 15 / 50
= **0.30**

P(D|B) = 25 / 50
= **0.50**

Step 3 — Total probability

P(D) = (3/4)(0.30) + (1/4)(0.50)
= 0.225 + 0.125
= **0.35**

Step 4 — Bayes' theorem

P(A|D) = [(3/4)(0.30)] / 0.35
= 0.225 / 0.35
= 225 / 350
= **9/14**

Final Answer

P(A|D) = 9/14 = 64.29%.

Question 13

Full Question: In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150. The dataset contains 1000 samples per class, and 20% of the data is used for testing. (5 Marks) Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

Step 1 — Total and test samples

Total = 1000 + 1000
= **2000**

Test samples = 20/100 × 2000
= **400**

Equal class distribution → positive test samples = **200**
Negative test samples = **200**

Step 2 — Find FN and FP

FN = Positive test samples − TP
= 200 − 120
= **80**

FP = Negative test samples − TN
= 200 – 150
= **50**

Step 3 — Accuracy

Accuracy = (TP + TN) / Total test samples
= (120 + 150) / 400
= 270 / 400
= **67.50%**

Step 4 — Precision

Precision = TP / (TP + FP)
= 120 / (120 + 50)
= 120 / 170
= **70.59%**

Step 5 — Sensitivity

Sensitivity = TP / (TP + FN)
= 120 / (120 + 80)
= 120 / 200
= **60.00%**

Step 6 — Specificity

Specificity = TN / (TN + FP)
= 150 / (150 + 50)
= 150 / 200
= **75.00%**

Question 14

Full Question: A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data. Calculate: Accuracy Precision Sensitivity Specificity. (5 Marks)

Step 1 — Total and test samples

Total = 1200 + 1200
= **2400**

Test samples = 25/100 × 2400
= **600**

Positive test samples = 600/2
= **300**

Negative test samples = 600/2
= **300**

Step 2 —Find FN and FP

FN = 300 − 180
= **120**

FP = 300 − 160
= **140**

Step 3 — Accuracy

Accuracy = (180 + 160) / 600
= 340 / 600
= **56.67%**

Step 4 — Precision

Precision = 180 / (180 + 140)
= 180 / 320
= **56.25%**

Step 5 — Sensitivity

Sensitivity = 180 / (180 + 120)
= 180 / 300
= **60.00%**

Step 6 — Specificity

Specificity = 160 / (160 + 140)
= 160 / 300
= **53.33%**

Question 15

Full Question: A deep learning model is trained for a classification task, and the following loss values are observed: (5 Marks) i. Epoch 1: Training Loss = 2.8, Validation Loss = 2.6; ii. Epoch 2: Training Loss = 2.4, Validation Loss = 2.2; iii. Epoch 3: Training Loss = 2.1, Validation Loss = 1.9; iv. Epoch 4: Training Loss = 1.9, Validation Loss = 1.8; v. Epoch 5: Training Loss = 1.7, Validation Loss = 1.9; vi. Epoch 6: Training Loss = 1.5, Validation Loss = 2.1; vii. Epoch 7: Training Loss = 1.3, Validation Loss = 2.4; viii. Epoch 8: Training Loss = 1.2, Validation Loss = 2.7. Answer the following: a. At which epoch does the model achieve the best generalization performance? b. At which point does overfitting begin? Justify your answer. c. Suggest two techniques to reduce overfitting and briefly explain how they help.

Step 1 — Find minimum validation loss

Validation loss:
Epoch 1 = 2.6
Epoch 2 = 2.2
Epoch 3 = 1.9
Epoch 4 = **1.8**
Epoch 5 = 1.9
Epoch 6 = 2.1
Epoch 7 = 2.4
Epoch 8 = 2.7

Step 2 — Answer (a)

Minimum validation loss = **1.8 at Epoch 4**.
Therefore, **best generalization performance occurs at Epoch 4**.

Step 3 — Answer (b)

At Epoch 4:
Training loss = 1.9
Validation loss = 1.8

At Epoch 5:
Training loss = 1.7 ↓
Validation loss = 1.9 ↑

Therefore, **overfitting begins at Epoch 5**.

Step 4 — Answer (c)

1. **Early stopping:** stop training when validation loss stops improving.
2. **Dropout / L2 regularization:** reduces model complexity and discourages memorization of training-specific patterns.

Final Answer

Best generalization = Epoch 4.
Overfitting begins = Epoch 5.
Techniques = Early stopping and Dropout/L2 regularization.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with P minor errors	Partially correct with In noticeable errors	correct or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts Lim with minor mistakes	ited or partially No or i correct application	ncorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some Inc parts	omplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-Un organized answer	derstandable with minor issues	Somewhat unclear or Difficult poorly structured	to understand